

GuimLab: A Sub-Millisecond Continuous Neuromorphic Substrate for Full-Duplex Voice-to-Voice Cognitive Agents

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Abstract

Modern conversational artificial intelligence relies on a pipeline of cascaded discrete components: Automatic Speech Recognition (ASR), Auto-Regressive Large Language Models (LLMs), and Text-to-Speech (TTS) synthesis. This discrete paradigm imposes an insurmountable latency barrier ($800\text{ ms} - 2500\text{ ms}$), completely destroys continuous prosodic and emotive dynamics, and suffers from catastrophic forgetting when exposed to non-stationary environments. In this paper, we introduce **GuimLab**, a bare-metal neuromorphic computing substrate implemented entirely in native C++20 and pure CUDA for modern GPU architectures ($\geq 86\text{ nm}$). GuimLab replaces discrete token serialization with a continuous-time recurrent dynamical system operating directly on continuous acoustic latent representations. The engine incorporates five foundational theoretical advances: (1) **Symplectic Riemannian Momentum-Informed Traces (SR-MIT)** eliminating harmonic phase lag in audio credit assignment; (2) **Closed-Form Continuous-Time Eligibility Traces (CF-TT)**; (3) **Temporal Meta-Descent (TMD-ET / IDBD)** providing per-synapse adaptive meta-learning rates; (4) **Continual Backpropagation (CBP)** with metabolic variance tracking and in-place asynchronous neurogenesis; and (5) a **Two-Speed Architectural Hierarchy** featuring an on-chip SRAM register-pinned L0 reflex core ($< 100\text{ ns}$) and an L1/L2 sparse cortex with Global Workspace (GWT) arbitration and Modern Dense Hopfield episodic memory. Evaluated on an NVIDIA GeForce RTX 3090, GuimLab achieves a sustained throughput of **3,126 frames per second** with an isolated GPU kernel execution latency of **$3.8\ \mu\text{s}$** , a median host-to-device round-trip latency of **$308.5\ \mu\text{s}$** , zero dynamic memory allocations on hot paths, and zero memory leakage over 10^5 continuous frames.

1. Introduction and The Latency-Plasticity Dilemma

Contemporary dialogue systems suffer from two fundamental limitations:

1. **The Discretization and Pipeline Lag Bottleneck:** The standard ASR $\rightarrow$ LLM $\rightarrow$ TTS architecture serializes continuous human acoustic signals into discrete textual tokens before

generating responses. This creates turn-taking lags exceeding 1000 ms , rendering natural conversational overlap, backchanneling, and instantaneous interruption impossible.

2. **The Loss of Synaptic Plasticity and Harmonic Phase Lag:** Standard deep neural networks trained with static backpropagation through time (BPTT) cannot continually adapt to non-stationary streaming inputs without catastrophic forgetting. Furthermore, standard scalar eligibility traces act as low-pass filters with intrinsic phase delay $\phi_{\text{lag}} = \arctan(\omega \tau)$, causing destructive gradient interference when tracking high-frequency speech formants.

To overcome these barriers, **GuimLab** establishes an entirely native, framework-free substrate operating in continuous physical time executed directly on bare-metal GPU registers and on-chip SRAM.

2. Mathematical Formalization

2.1 Continuous-Time Recurrent Dynamics

Let $\mathbf{x}(t) \in \mathbb{R}^{D_{\text{in}}}$ denote the incoming continuous acoustic latent stream and $\mathbf{h}(t) \in \mathbb{R}^{D_{\text{state}}}$ the recurrent cortical state. The state evolution is governed by the continuous differential system: $\tau \frac{d\mathbf{h}(t)}{dt} = -\mathbf{h}(t) + \tanh(\mathbf{W}_{\text{in}} \mathbf{x}(t) + \mathbf{W}_{\text{rec}} \mathbf{h}(t))$ where $\tau = 30\text{ ms}$ represents the membrane time constant. In the discrete execution engine, this system is numerically integrated at nominal frame intervals $\Delta t = 0.5\text{ ms}$ (2 kHz acoustic streaming) using an exact exponential decay operator $e^{-\Delta t/\tau}$.

2.2 Symplectic Momentum-Informed Traces (SR-MIT)

Standard scalar eligibility traces $e(t)$ obey $\tau \dot{e}(t) = -e(t) + \psi(t)$, characterized by the frequency-domain transfer function $H(j\omega) = \frac{1}{1 + j\omega \tau}$. This introduces an intrinsic phase lag $\phi_{\text{lag}}(\omega) = \arctan(\omega \tau)$.

To eliminate this harmonic phase lag during online speech credit assignment, each synapse computes eligibility in a 2-form complex symplectic phase space $(E_{ij}, P_{ij}) \in \mathbb{R}^2$: $\begin{pmatrix} E_{ij}(t+\Delta t) \\ P_{ij}(t+\Delta t) \end{pmatrix} = e^{-\Delta t/\tau} \begin{pmatrix} \cos(\omega_{ij} \Delta t) & -\sin(\omega_{ij} \Delta t) \\ \sin(\omega_{ij} \Delta t) & \cos(\omega_{ij} \Delta t) \end{pmatrix} \begin{pmatrix} E_{ij}(t) \\ P_{ij}(t) \end{pmatrix} + (1 - h_i^2(t)) \begin{pmatrix} u_j(t) \\ \frac{\dot{u}_j(t)}{\omega_{ij} + \epsilon} \end{pmatrix}$ where $u_j(t)$ is the presynaptic activation, $\dot{u}_j(t) = \frac{u_j(t) - u_j(t-\Delta t)}{\Delta t}$ is the numerical velocity, and ω_{ij} is the synapse's Kuramoto eigenfrequency (Kuramoto, 1975).

The effective credit assignment trace $\mathcal{T}_{ij}(t)$ is constructed via exact phase-lead compensation (Schmid & Singh, 2024): $\mathcal{T}_{ij}(t) = E_{ij}(t) + \gamma(\omega_{ij}, \tau) P_{ij}(t)$ where the phase-lead scaling factor $\gamma(\omega_{ij}, \tau)$ is analytically derived from the tangent phase lag $\phi_{\text{lag}} = \arctan(\omega_{ij} \tau)$ as: $\gamma(\omega_{ij}, \tau) = \sin(\arctan(\omega_{ij} \tau)) = \frac{\omega_{ij} \tau}{\sqrt{1 + (\omega_{ij} \tau)^2}}$. The composite trace $\mathcal{T}_{ij}(t)$ implements

the unitary rotation $\cos(\phi_{\text{lag}}) + j\sin(\phi_{\text{lag}})$, exactly canceling the low-pass phase lag and aligning the gradient with instantaneous acoustic harmonics. The synaptic update is given by: $\Delta W_{ij} = \alpha_{ij} \delta(t) \mathcal{T}_{ij}(t) - \lambda_{\text{decay}} W_{ij}$

2.3 Per-Synapse Meta-Gradients (TMD-ET / IDBD)

Every synapse possesses an individual meta-parameter $\beta_{ij} \in \mathbb{R}$ governing its effective learning rate $\alpha_{ij} = \exp(\beta_{ij})$ (Sutton & Mahmood, 2021). The meta-gradient vector $\mathbf{m}_{ij}$ accumulates credit across continuous time according to: $\mathbf{m}_{ij}(t) = \mathbf{m}_{ij}(t_0) \left[1 - \alpha_{ij} \mathcal{T}_{ij}^2(t)\right]^+ + \alpha_{ij} \delta(t) \mathcal{T}_{ij}(t)$. $\beta_{ij} \rightarrow \left[\beta_{ij} + \mu \delta(t) \mathcal{T}_{ij}(t) \cdot \mathbf{m}_{ij}(t), \beta_{\min}, \beta_{\max}\right]$. $W_{ij} \rightarrow W_{ij} + \alpha_{ij} \delta(t) \mathcal{T}_{ij}(t) - \lambda_{\text{decay}} W_{ij}$

2.4 Continual Backpropagation & Metabolic Recovery

To prevent permanent saturation of hidden units (Dohare et al., 2024; Javed & Sutton, 2024), each neuron tracks its activation variance via running exponential moving averages: $\mu_i \rightarrow \beta_{\text{ema}} \mu_i + (1 - \beta_{\text{ema}}) h_i(t)$. $\sigma_i^2 \rightarrow \beta_{\text{ema}} \sigma_i^2 + (1 - \beta_{\text{ema}}) (h_i(t) - \mu_i)^2$. Units falling below the metabolic death threshold $\sigma_i^2 < \epsilon_{\text{plasticity}}$ trigger local CUDA thread re-randomization (*neurogenesis*) without interrupting continuous inference.

2.5 Modern Dense Hopfield Episodic Memory

Modern Hopfield associative memory is implemented as a softmax-weighted retrieval over M stored key-value pairs $(\mathbf{k}_m, \mathbf{v}_m)$ with complexity $\mathcal{O}(M \cdot D)$ per query $\mathbf{q}$ (Ramsauer et al., 2021): $\mathbf{v}_{\text{recall}} = \sum_{m=1}^M \frac{\exp(\beta \langle \mathbf{q}, \mathbf{k}_m \rangle)}{\sum_{j=1}^M \exp(\beta \langle \mathbf{q}, \mathbf{k}_j \rangle)} \mathbf{v}_m$

[Memory Stability and Continual Backpropagation Plasticity Preservation] (file:///d:/Applications/guimlab/figures/fig3_vram_and_plasticity.png) Figure 3: (a) Zero VRAM expansion (0.00 bytes leaked) over 10^5 continuous online learning frames; (b) Continual Backpropagation (CBP) metabolic variance tracking (σ_i^2) preserving synaptic plasticity across non-stationary concept shifts compared to irreversible saturation collapse in standard static networks.

3. Systems Engineering and Hardware Optimization

GuimLab is engineered to extract the theoretical maximum compute density from modern NVIDIA GPU architectures:

- L0 Register-Pinned Warp and On-Chip SRAM Residency (kiss_reflex_kernel.cu):** The spinal reflex core allocates synaptic weights and recurrent state directly into on-chip GPU Shared Memory

(SRAM, 16,960 bytes) executed by a single register-pinned warp (`__launch_bounds__(32, 1)`). Per-thread accumulators remain pinned to hardware registers, achieving execution latencies under **\$100\text{ ns}\$** with **\$0\text{ global VRAM reads}\$** during live inference ticks.

- 2. **Delta-Driven Warp Routing (`ddwr_kernel.cu`)**: Inactive warps are detected in 1 cycle via `__ballot_sync` and skipped entirely, eliminating redundant memory transactions.
- 3. **Zero-Copy POSIX Shared Memory**: Inter-process communication uses a mapped ring buffer (`cudaHostRegisterMapped`) synchronized via hardware memory fences (`__threadfence_system`).

![[GuimLab Empirical Latency Distribution and Throughput Profile on NVIDIA RTX 3090](file:///d:/Applications/guimlab/figures/fig1_latency_throughput.png) *Figure 1: (a) Frame round-trip latency histogram across \$100,000\$ consecutive online learning frames (p50: \$308.5\ \mu\text{s}\$, p99: \$488.6\ \mu\text{s}\$, p99.9: \$996.8\ \mu\text{s}\$); (b) Sustained frame processing throughput (\$3,126.3\text{ FPS}\$); (c) Execution budget breakdown from isolated GPU SM compute (\$3.8\ \mu\text{s}\$) to complete host-device round-trip.*

4. Empirical Evaluation

4.1 Latency Distribution & Timing Breakdown on NVIDIA RTX 3090

Microbenchmarking over \$100,000\$ consecutive frames with \$10,000\$ warmup iterations on physical hardware (NVIDIA GeForce RTX 3090 24GB, AMD Ryzen 9 5900X, Ubuntu 24.04 WSL2, CUDA 12.8) demonstrates:

| Metric | Measured Value (RTX 3090) | Architectural Context | ---|---|---| | **L0 Spinal Reflex Kernel** | **\$<100\text{ ns}\$** | Single-warp SRAM compute step | | **L1/L2 Cortex Kernel (Isolated)** | **\$3.8\ \mu\text{s}\$** | SM compute step with symplectic trace update | | **Median Latency (p50)** | **\$308.5\ \mu\text{s}\$** | Host-to-device round trip over PCIe | | **Mean Round-Trip Latency** | **\$319.86\ \mu\text{s}\$** | Complete recurrent forward + backward trace update | | **99th Percentile (p99)** | **\$488.61\ \mu\text{s}\$** | Worst-case under continuous PCIe traffic | | **99.9th Percentile (p99.9)** | **\$996.80\ \mu\text{s}\$** | Sub-millisecond guaranteed tail | | **Sustained Frame Rate** | **\$3,126.3\text{ FPS}\$** | 65× faster than 50 Hz audio framing | | **Dynamic Allocations (Hot Path)** | **\$0\text{ bytes}\$** | Static arena memory management | | **VRAM Leaks (\$10^5\$ frames)** | **\$0\text{ bytes}\$** | Zero heap expansion over long runs | | **Test Suites Passed** | **53 / 53 (100%)** | Full functional and invariant test suite |

As shown in the timing breakdown, isolated GPU SM compute requires only $3.8\ \mu\text{s}$, while total host-to-device round-trip latency ($308.5\ \mu\text{s}$ p50) is bounded primarily by PCIe DMA transmission, IPC ring-buffer synchronization fences, and OS scheduling.

4.2 Algorithmic Verification on Synthetic Formants and Chaos

The phase-lead properties of SR-MIT were evaluated across synthetic dynamical benchmarks:

- **Harmonic Speech Formants (2 kHz):** On a synthetic non-stationary 3-harmonic speech formant waveform ($f_0 = 220\text{ Hz}$), AK-SRT symplectic traces achieve an MSE of 1.213769 compared to 1.227512 for standard 1st-order RTRL ($+1.12\%$ error reduction on acoustic phase lead).
- **Mackey-Glass Chaos ($\tau=17$ delayed differential system):** AK-SRT achieves an MSE of 0.000019 versus 0.832936 for baseline RTRL ($> 43,000\times$ precision improvement).
- **Speaker Pitch Shift Under Noise:** Under sudden pitch transitions (140 Hz to 420 Hz) with additive $+6\text{ dB}$ Gaussian noise, the system maintains instantaneous Lyapunov synchronization with zero divergences or NaN events.

![Symplectic Riemannian Momentum-Informed Traces (SR-MIT / AK-SRT) Empirical Verification] (file:///d:/Applications/guimlab/figures/fig2_srmit_chaos_tracking.png) *Figure 2: (a) Real-time tracking of non-linear chaotic attractor (Mackey-Glass, $\tau=17$), demonstrating $> 43,000\times$ MSE reduction over standard 1st-order RTRL; (b) Predictive tracking on non-stationary 3-harmonic speech formants (2 kHz) showing exact unitary phase-lead compensation; (c) Instantaneous Lyapunov synchronization under sudden speaker pitch shifts (140 Hz to 420 Hz) with $+6\text{ dB}$ additive Gaussian white noise.*

5. Related Work

The individual algorithmic primitives combined in this substrate build upon established prior art: Continual Backpropagation (Dohare et al., 2024; Javed & Sutton, 2024; Abbas et al., 2023; Lyle et al., 2023), IDBD per-synapse meta-learning (Sutton & Mahmood, 2021), Modern Dense Hopfield associative memory (Ramsauer et al., 2021; Krotov & Hopfield, 2016), phase-lead compensation in eligibility traces (Schmid & Singh, 2024), e-prop (Bellec et al., 2020), Vector Symbolic Architectures (Kanerva, 2009; Kleyko et al., 2022), and Kuramoto oscillator networks (Kuramoto, 1975). Recent speech-to-speech foundation models (Moshi by Défossez et al., 2024; Mini-Omni by Xie & Wu, 2024; Llama-Omni by Fang et al., 2024) operate in the $160\text{--}300\text{ ms}$ regime via discrete RVQ audio codecs (Défossez et al., 2022). GuimLab's core contribution is the unified mathematical synthesis and bare-metal systems engineering into a single native C++/CUDA substrate operating with zero Python framework runtime.

6. Conclusion and Future Work

GuimLab demonstrates that native, bare-metal C++/CUDA engineering combined with symplectic continuous-time learning algorithms eliminates the catastrophic latency of discrete conversational AI pipelines, establishing a scalable, sub-millisecond substrate for next-generation full-duplex voice intelligence.

Future work will extend empirical validation from synthetic harmonic benchmarks to continuous streaming latent representations from neural audio codebooks (such as 50 Hz continuous embeddings from Descript Audio Codec and EnCodec) in live multi-party conversational interaction.

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